Integrated Rate Laws and Reaction Mechanisms

Answer key with worked solutions

A. Understanding

1. Definitions.

> (a) **Half-life:** the time for the concentration of a reactant to fall to half its value. (b) **Elementary step:** a single step in a mechanism, whose rate law follows from the particles that collide. (c) **Rate-determining step:** the **slowest** step, which sets the overall rate. (d) **Intermediate:** a species **made** in one step and **used up** in a later one.

2. Plots and units.

> **Zero:** [A] against t, slope −k, k in $\mathbf{M\ s^{-1}}$. **First:** ln[A] against t, slope −k, k in $\mathbf{s^{-1}}$. **Second:** 1/[A] against t, slope +k, k in $\mathbf{M^{-1}\ s^{-1}}$.

B. Calculations

3. Drug elimination.

> (a) $k = \dfrac{\ln 2}{t_{1/2}} = \dfrac{0.6931}{4.00\ \text{h}} = \mathbf{0.173\ h^{-1}}$
>
> (b) $n = \dfrac{10.0\ \text{h}}{4.00\ \text{h}} = 2.50$ half-lives; $200.\ \text{mg} \times \left(\tfrac{1}{2}\right)^{2.50} = \mathbf{35.4\ mg}$

4. First-order time.

> $t = \dfrac{\ln([\text{A}]_0/[\text{A}])}{k} = \dfrac{\ln(0.800\ \text{M}/0.200\ \text{M})}{0.0200\ \text{min}^{-1}} = \dfrac{1.386}{0.0200\ \text{min}^{-1}} = \mathbf{69.3\ min}$ (exactly two half-lives of 34.7 min)

5. Second order.

> (a) $t_{1/2} = \dfrac{1}{(0.150\ \text{M}^{-1}\,\text{s}^{-1})(0.0500\ \text{M})} = \mathbf{133\ s}$
>
> (b) $\dfrac{1}{[\text{A}]} = \dfrac{1}{0.0500\ \text{M}} + (0.150\ \text{M}^{-1}\,\text{s}^{-1})(200.\ \text{s}) = 20.0\ \text{M}^{-1} + 30.0\ \text{M}^{-1} = 50.0\ \text{M}^{-1}$, so $[\text{A}] =$ **0.0200 M**

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6. Zero order.

> (a) $t_{1/2} = \dfrac{0.500\ \text{M}}{2(2.50 \times 10^{-3}\ \text{M/s})} = \mathbf{100.\ s}$ (b) $t = \dfrac{0.500\ \text{M}}{2.50 \times 10^{-3}\ \text{M/s}} = \mathbf{200.\ s}$

C. Order from data

7. First order from data.

> In each 100 s interval [A] is multiplied by the **same fraction**: $\dfrac{0.300}{0.400} = 0.750$, $\dfrac{0.225}{0.300} = 0.750$, $\dfrac{0.169}{0.225} = 0.751$. A constant fraction in equal times means **first order** (ln[A] against t is a straight line).
> $k = \dfrac{\ln(0.400\ \text{M}/0.300\ \text{M})}{100\ \text{s}} = \dfrac{0.2877}{100\ \text{s}} = \mathbf{2.88 \times 10^{-3}\ s^{-1}}$

D. Mechanisms

8. Mechanism.

> (a) Add the steps and cancel Cl: $2\,NO_2Cl \longrightarrow 2\,NO_2 + Cl_2$ (b) **Cl** (made in step 1, used in step 2) (c) The slow step has one NO_2Cl: **rate = k[NO2Cl]**

E. Challenge

9. Successive half-lives.

> First: $t_{1/2} = \dfrac{1}{(0.0500\ \text{M}^{-1}\,\text{s}^{-1})(0.100\ \text{M})} = \mathbf{200.\ s}$. Second: $t_{1/2} = \dfrac{1}{(0.0500\ \text{M}^{-1}\,\text{s}^{-1})(0.0500\ \text{M})} = \mathbf{400.\ s}$
> Each half-life is **twice** the one before. If measured half-lives keep doubling, the reaction is second order (first order gives equal half-lives).

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